Energy Monitoring System – Full Stack Development

Project Overview

This project involved the complete design and development of a **Full Stack Energy Monitoring System (EMS)** as part of an industry internship at a multinational technology organization. The system was developed to enable efficient monitoring, visualization, and analysis of energy consumption data through an intuitive and interactive dashboard.

In addition to frontend development, I also developed the **entire backend system using the Django framework**, implemented a **SQL-based relational database**, and handled the **complete integration between frontend and backend**, making it a fully functional end-to-end project.

The system presents complex energy metrics in a structured and user-friendly format, allowing users to analyze energy usage patterns, simulate billing scenarios, and gain cost-related insights effectively.

Role & Responsibilities

- Designed and developed responsive frontend components for the Energy Monitoring System
- Built interactive dashboards to visualize energy consumption trends and key performance indicators
- Developed the complete backend using **Django**
- Designed and managed database schemas using **SQL**
- Created RESTful APIs to enable seamless frontend–backend communication
- Integrated frontend (React.js) with Django backend for real-time data exchange
- Implemented UI layouts for dashboards, usage breakdowns, and billing simulations
- Ensured application performance, scalability, and optimized database queries
- Followed industry-standard development practices in a production-oriented environment

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Key Features Implemented

- Energy consumption visualization dashboards
- Interactive charts and trend analysis graphs
- User-based energy data management
- Billing simulations and cost breakdown modules
- Secure backend data handling and API management
- Complete frontend–backend integration

- Modular and scalable full-stack architecture
- Fully responsive design for enhanced user experience

Technologies & Tools

Frontend

- React.js
- Responsive UI Design
- Data-driven charting libraries

Backend

- Django Framework
- SQL (Relational Database Management System)
- REST API Development

Integration & Architecture

- Frontend–Backend API Integration
- Database Design & Management
- End-to-End Full Stack Implementation